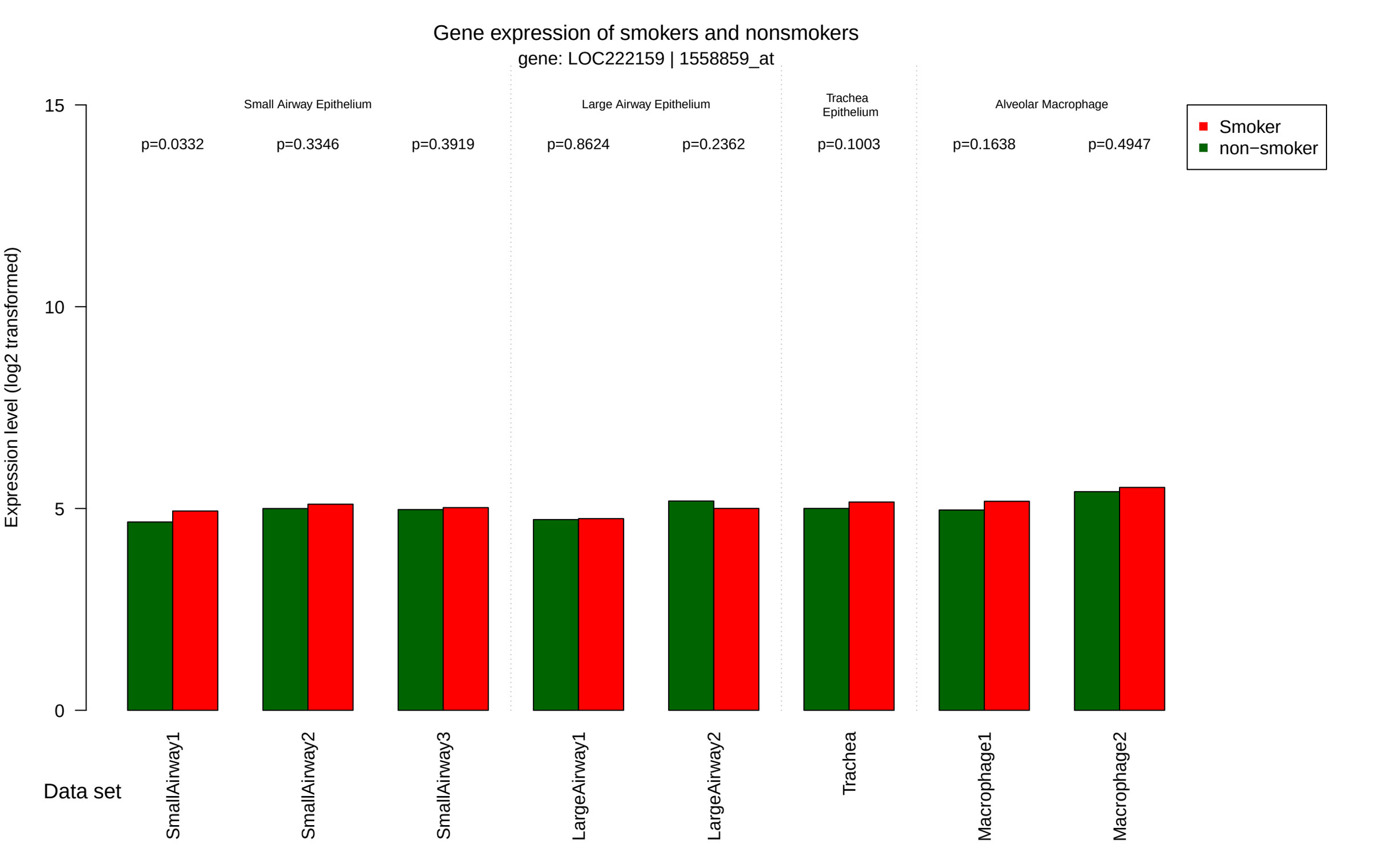
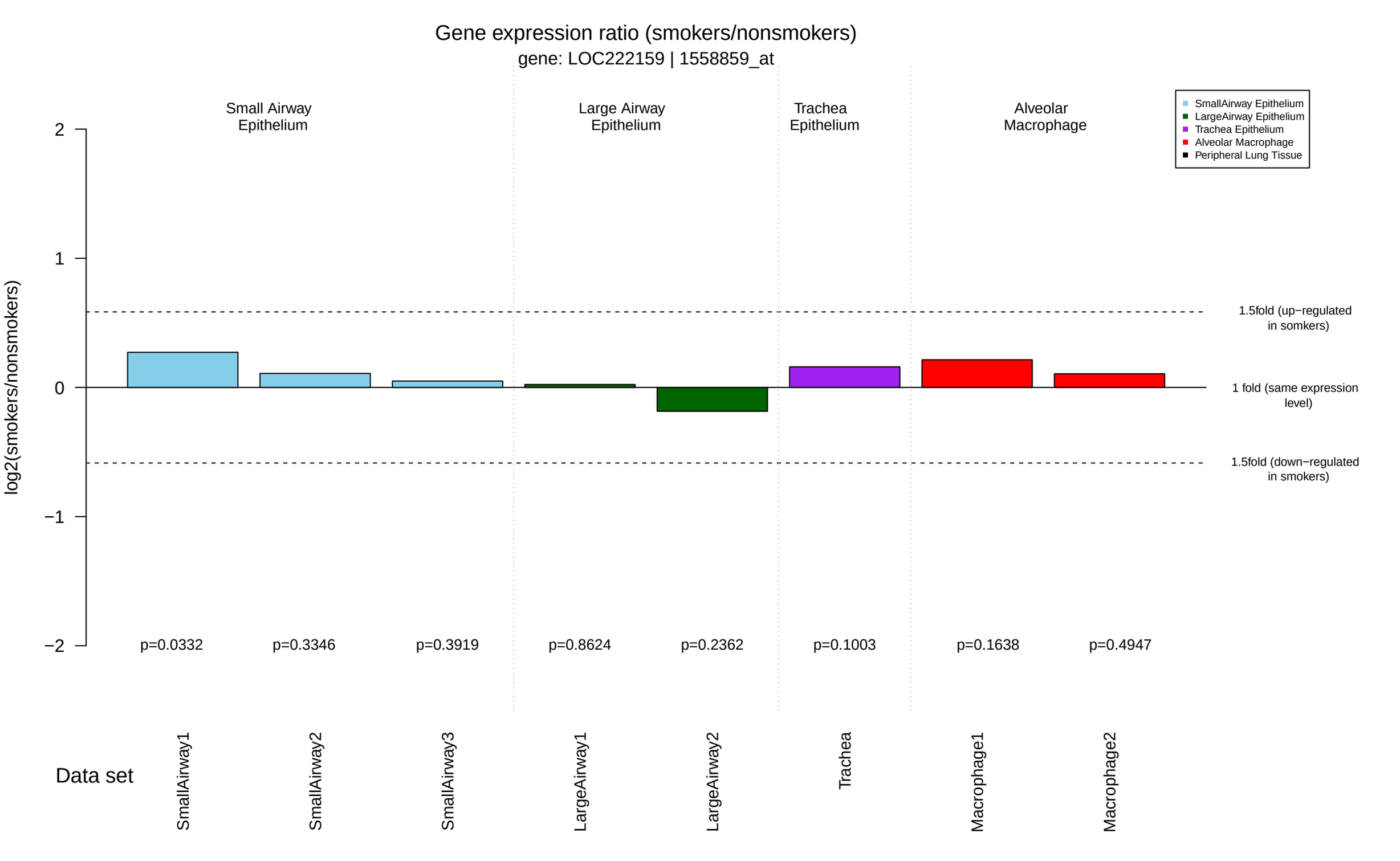
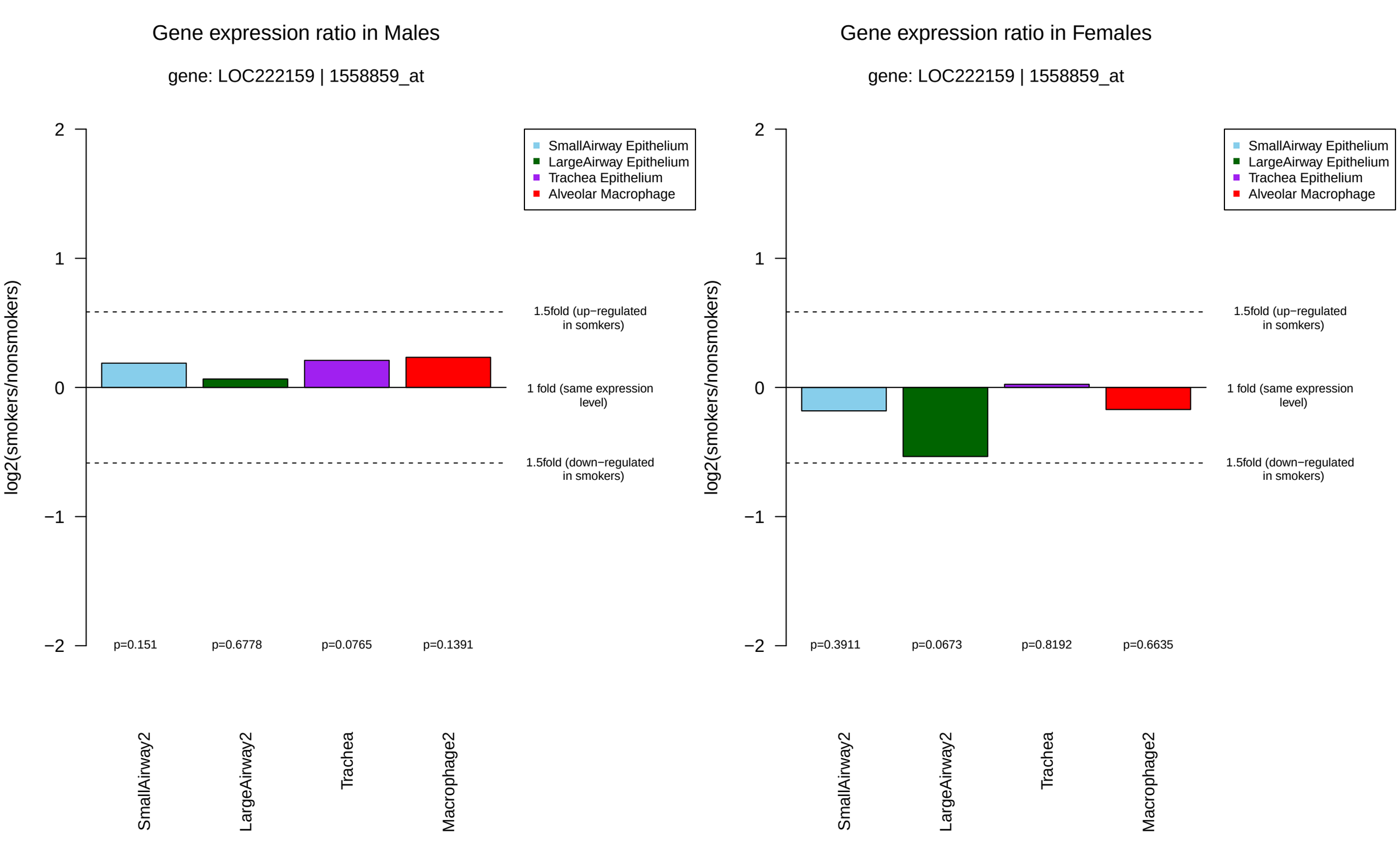
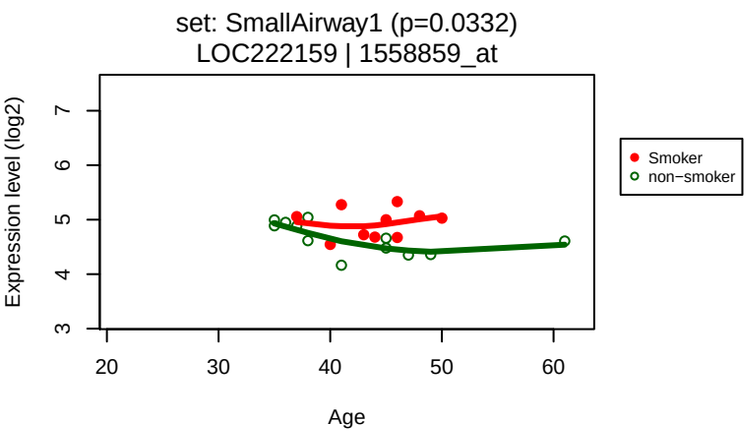
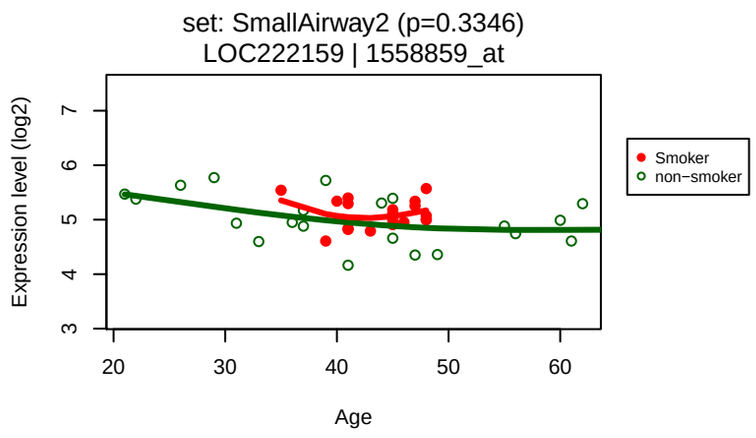
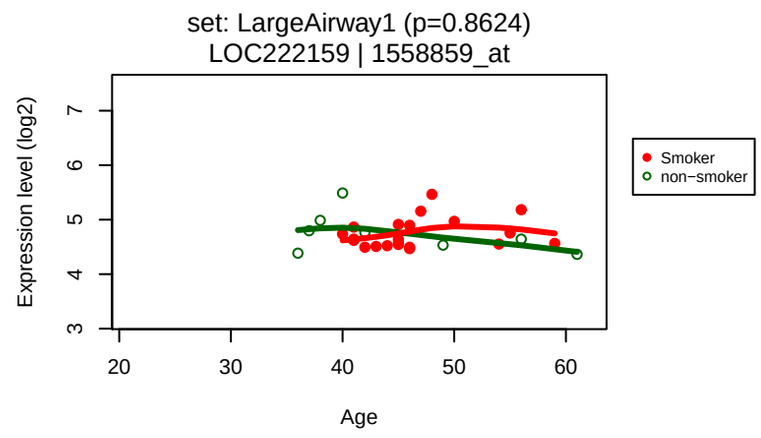

Gene expression of smokers and nonsmokers
gene: LOC222159 | 1558859_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0332
p=0.3346
p=0.3919
p=0.8624
p=0.2362
p=0.1003
p=0.1638
p=0.4947
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

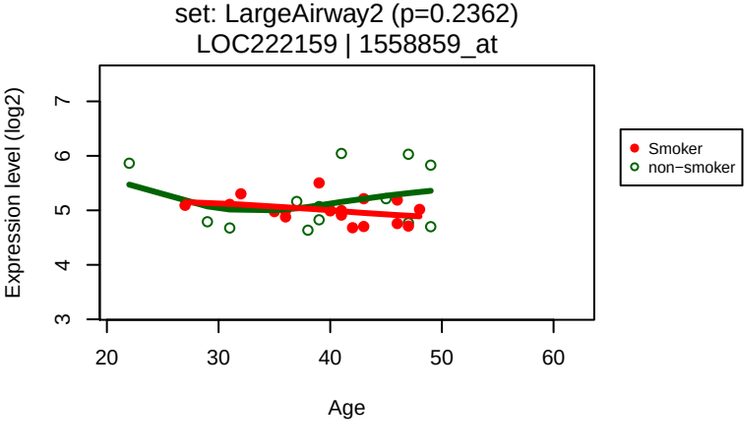

Gene expression ratio (smokers/nonsmokers)
gene: LOC222159 | 1558859_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0332
p=0.3346
p=0.3919
p=0.8624
p=0.2362
p=0.1003
p=0.1638
p=0.4947
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

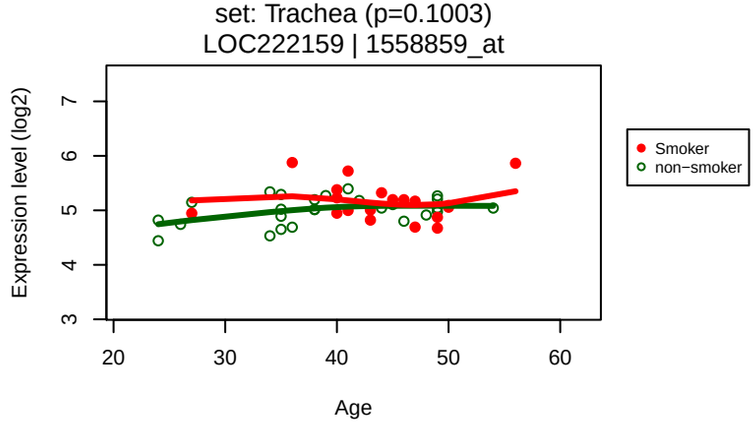

Gene expression ratio in Males
gene: LOC222159 | 1558859_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.151
p=0.6778
p=0.0765
p=0.1391
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC222159 | 1558859_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3911
p=0.0673
p=0.8192
p=0.6635
SmallAirway2
LargeAirway2
Trachea
Macrophage2

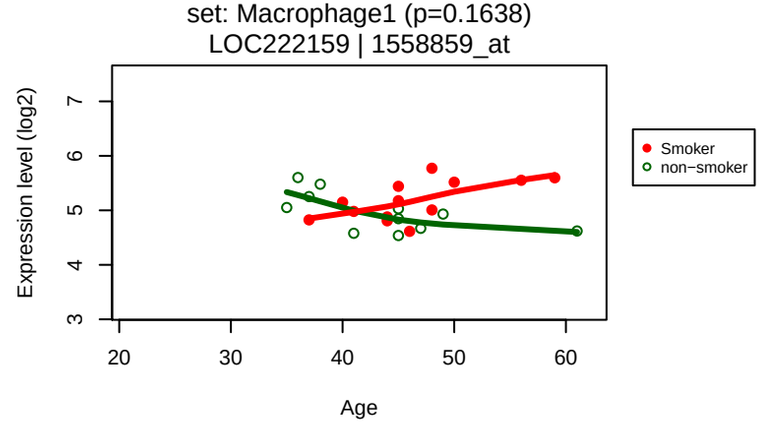

set: SmallAirway1 (p=0.0332)
LOC222159 | 1558859_at
Expression level (log2)
Age
Smoker
non−smoker

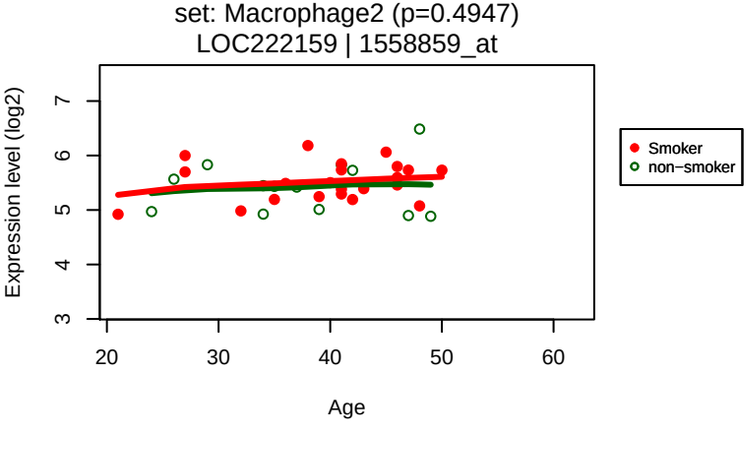

set: SmallAirway2 (p=0.3346)
LOC222159 | 1558859_at
Expression level (log2)
Age
Smoker
non−smoker

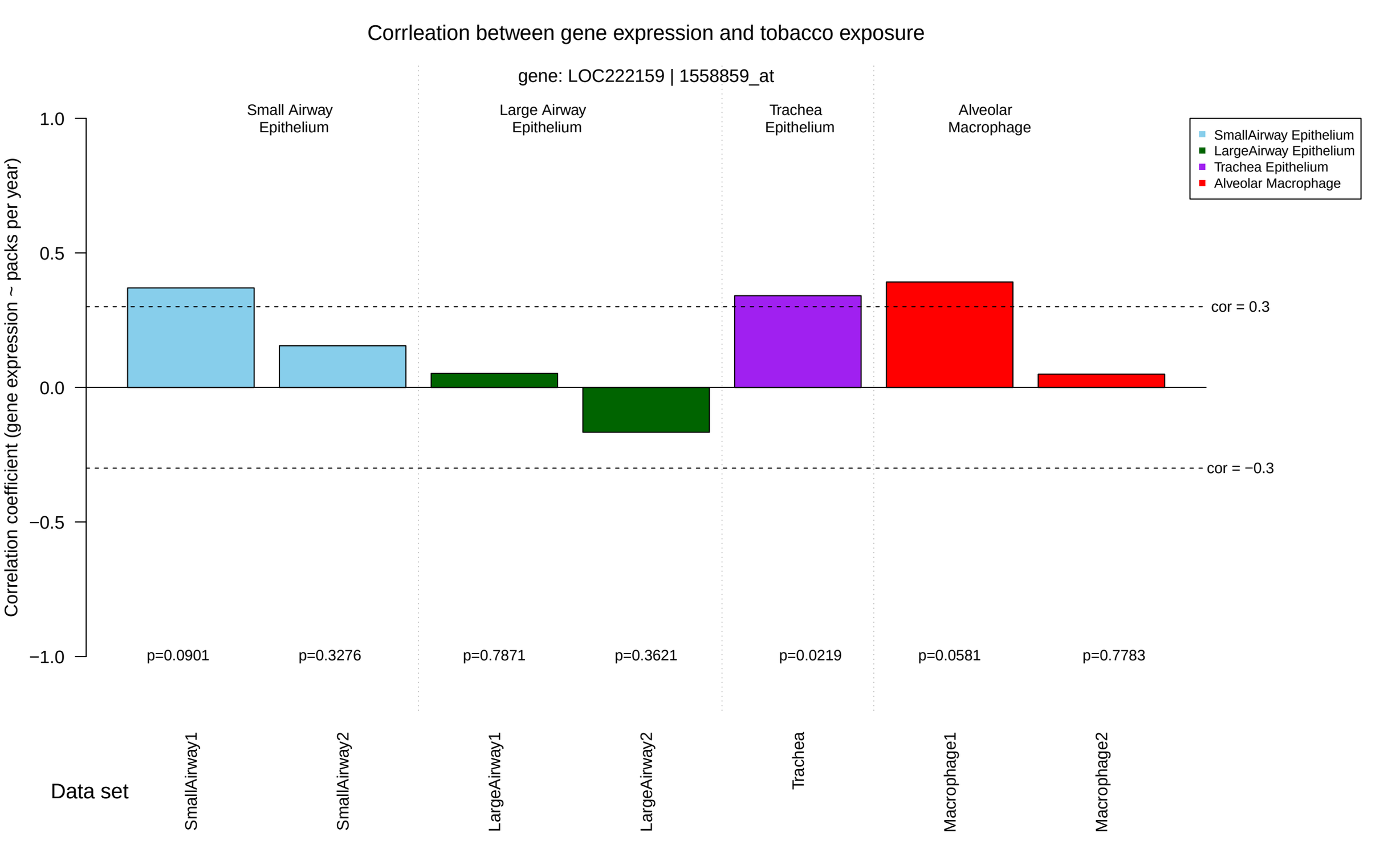

set: LargeAirway1 (p=0.8624)
LOC222159 | 1558859_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2362)
LOC222159 | 1558859_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1003)
LOC222159 | 1558859_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1638)
LOC222159 | 1558859_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4947)
LOC222159 | 1558859_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC222159 | 1558859_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0901
p=0.3276
p=0.7871
p=0.3621
p=0.0219
p=0.0581
p=0.7783
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2